

Externalities in Economies with Imperfect Information and Incomplete Markets

A Complete Analytical Walkthrough

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Introduction and Core Thesis

- **Traditional View:** Pecuniary externalities merely transfer income via price changes and net out to zero at the margin.
- **The Core Breakthrough:** When information is imperfect or markets are incomplete, pecuniary externalities do **not** net out.
- **Systemic Inefficiency:** Competitive market equilibria are rarely constrained Pareto efficient.
- **Policy Implication:** Government tax/subsidy interventions can virtually always achieve a structural Pareto improvement.

The General Model: Households

- Households ($h = 1, \dots, H$) maximize utility:

$$u^h(x^h, z^h)$$

- $x^h = (x_1^h, \bar{x}^h)$: x_1^h is numeraire, $\bar{x}^h$ represents nonnumeraire goods.
- z^h : Informational/externality variables taken as fixed by households.
- **Budget Constraint** (q is consumer price vector, I^h is transfer):

$$x_1^h + q \cdot \bar{x}^h \leq I^h + \sum_f a^{hf} \cdot \pi^f$$

- **Expenditure Function:** $E^h(q, z^h, u^h)$ where $\frac{\partial E^h}{\partial q_k} = \hat{x}_k^h$.

The General Model: Firms and Government

- Firms maximize profits ($\bar{y}^f$ nonnumeraire production vector):

$$\pi^f = y_1^f + p \cdot \bar{y}^f$$

- **Production Constraint:** $y_1^f - G^f(\bar{y}^f, z^f) \leq 0$.
- **Envelope Property:** $\frac{\partial \pi_*^f}{\partial p_k} = y_k^f$ (supply function).
- Government sets taxes $t \equiv (q - p)$ and tracks net revenue R :

$$R \equiv t \cdot \bar{x} - \sum_H I^h$$

The General Efficiency Test Framework

- Test efficiency by checking if a small tax vector dt can increase government revenue R while keeping utilities $\bar{u}^h$ constant.
- Differentiating the budget envelope yields total required transfers:

$$\sum_H \frac{dl^h}{dt} = \bar{x} - \left(\sum_F \pi_z^f \frac{dz^f}{dt} - \sum_H E_z^h \frac{dz^h}{dt} \right)$$

- Marginal effect of taxes on government revenue:

$$\frac{dR}{dt} = \frac{d\bar{x}}{dt} \cdot t + (\Pi^t - B^t)$$

- Where $\Pi^t \equiv \sum_F \pi_z^f \frac{dz^f}{dt}$ and $B^t \equiv \sum_H E_z^h \frac{dz^h}{dt}$.
- At $t = 0$, if $\frac{dR}{dt} = \Pi^t - B^t \neq 0$, the market is inefficient.

Application 1 & 2: Tax Distortions & Adverse Selection

- **Pre-existing Tax Distortions:** Rewrites standard distortions as a z externality. A small tax on a Hicks substitute improves welfare:

$$\left. \frac{\partial R}{\partial t_i} \right| = t_1 \left(\frac{dx_1}{dt_i} \right)_{\bar{u}}$$

- **Adverse Selection:** Uninformed buyers face average quality $\bar{\theta}$.

$$\frac{dR}{dt} = \left[\sum_F \pi_{\theta}^f - \sum_H E_{\theta}^h \right] \cdot \frac{d\bar{\theta}}{dt}$$

- **Mechanism:** Individual trades alter $\bar{\theta}$, meaning taxes shifting consumer choices to raise market quality are Pareto-improving.

Application 3: Signaling and Screening

- High-quality agents buy costly signals to separate themselves.
- Let n_1 be the fraction of agents signaling. The sorting value is:

$$\frac{dR}{dt} = \left[n_1 \frac{\partial \bar{\theta}_1}{\partial t} \right] \left[\frac{\partial \bar{y}_{01}}{\partial \bar{\theta}_1} - \frac{\partial \bar{y}_{02}}{\partial \bar{\theta}_2} \right] - \frac{\partial n_1}{\partial t} \left[\frac{\partial \bar{y}_{02}}{\partial \bar{\theta}_2} \right] (\bar{\theta}_1 - \bar{\theta}_2)$$

- Under pure hierarchical screening (no differential sorting value):

$$\frac{dR}{dt} = - \frac{\partial n_1}{\partial t} \left[\frac{\partial \bar{y}_{02}}{\partial \bar{\theta}_2} \right] (\bar{\theta}_1 - \bar{\theta}_2)$$

- **Policy:** Taxing signaling reduces the volume of socially wasteful over-education/sorting (n_1).

Application 4: Moral Hazard

- Effort e^h to prevent accidents is hidden from insurers.
- Individual insurance premiums depend on group average effort $\bar{e}$.
- Welfare impact of a tax intervention:

$$\frac{dR}{dt} = \sum_H \frac{d\epsilon^l(\mu^h|\bar{e})}{d\bar{e}} \cdot \frac{d\bar{e}}{dt}$$

- Because extra care reduces aggregate insurance payout demands, the derivative is structurally non-zero.
- **Policy:** Subsidize complements to care (fire extinguishers) and tax substitutes (alcohol) to artificially induce optimal effort.

Model Modification 1: Incomplete Risk Markets

- **Structural Shift:** Transitions to a 2-period model ($t = 1, 2$) with K states of nature in period 2.
- Good 0 acts as a store of value. Nonnumeraire relative prices in state k are represented by vector s_k .
- Expected Utility: $V^h = \sum_k b_k U^h(x_{1k}^h, x_k^h)$.
- Price vectors s_k act directly as the informational externalities.
- **Modified Welfare Impact Equation:**

$$\frac{dR}{dt} = \sum_k \left[\sum_H \tilde{x}_k^h \cdot \frac{\lambda_k^h}{U_1^h} \right] \frac{ds_k}{dt} b_k$$

- Where λ_k^h is marginal utility of income in state k . Changes in relative prices via taxation structurally rebalance risk-sharing pools.

Model Modification 2: Queue Rationing

- **Structural Shift:** A single good is supplied across N distinct markets, each defined by an average waiting time T_l .
- Consumer price is modified to a vector: $q_l = (p_l, T_l)$.
- Firms face customer arrival rates Ψ^f , which depend on their machine reliability r^f and average market reliability $\bar{r}$.
- **Modified Welfare Impact Equation:**

$$\frac{dR}{dt} = \sum_i \sum_{F_i} (p_i r^f - c_y^f) \left(\frac{\partial \Psi^f}{\partial \bar{y}_i} \cdot \frac{d\bar{y}_i}{dt} + \frac{\partial \Psi^f}{\partial \bar{r}_i} \cdot \frac{d\bar{r}_i}{dt} \right)$$

- **Mechanism:** Unreliable firms break down and dump arrivals onto other queues. Taxes optimize service capacity configuration.

Conclusion and Key Takeaways

- **The Paradigm Shift:** The Arrow-Debreu ideal is a mathematical anomaly; incomplete information and missing insurance are the structural reality.
- **Pecuniary Externalities Matter:** In the real world, price shifts alter unobservable vectors (z, s_k, T_l) , failing to net out.
- **Feasible Policy Design:** Corrective taxes do not require omniscience; they can be designed around observable macroeconomic targets.